Customer/Staff Fast Food Ordering System
Requirements Specification Document

1 Introduction

1.1 Purpose

The purpose of this system is to describe the functionality requirements for the food ordering system. The system identifies key features designed to focus on the ordering/managing process for both customers and employees.
This system also provides clarity on system features, interfaces, and performance expectations for developers. Should there be any questions of implementation, this shall serve as a reference which will then allow easy contact access for questions and answers.

1.2 Scope

The scope of the system is to allow proper management of orders by the staff members who have access to the system to update live tracking/orders. The system also has an ordering system that allows customers to view items, edit their cart, and push orders to the staff side.

1.3 Definitions

UI (User interface)
API (Application programming interface)
Cart (list of orderable items selected by customer)
Order Manager (managed by staff for customer orders)
Inventory Tracker (Tracks inventory of items)

1.4 References

4th Edition Phillip A. Laplante and Mohamad H. Kassab
https://instructure.charlotte.edu/courses/258585/files/31590369?module_item_id=7620125

1.5 Overview of SRS

This document describes requirements listed below.
Section 2 Overall Perspective

- Product perspective
- User Interfaces
- Use case diagram
- Hardware interfaces
- Software interfaces
- Communication interfaces
- Memory constraints
- Operation

- Product function
- User characteristics
- Constraints
- No budget
- Assumptions and dependencies
- Apportioning of requirements

Section 3 Specific Requirements

- External interfaces requirements
- Functional requirements
- Non-functional requirements

Section 4 Supporting Information

- Appendices
- Team contribution list

2 Overall Description

2.1 Product Perspective

This product is an online, web-based system that will be used to streamline the management of customer orders. From the orders being placed to being completed, the system will operate through up-to-date browsers. The system will combine front-end programming such as HTML, CSS, and JavaScript to deliver a seamless and intuitive user experience.

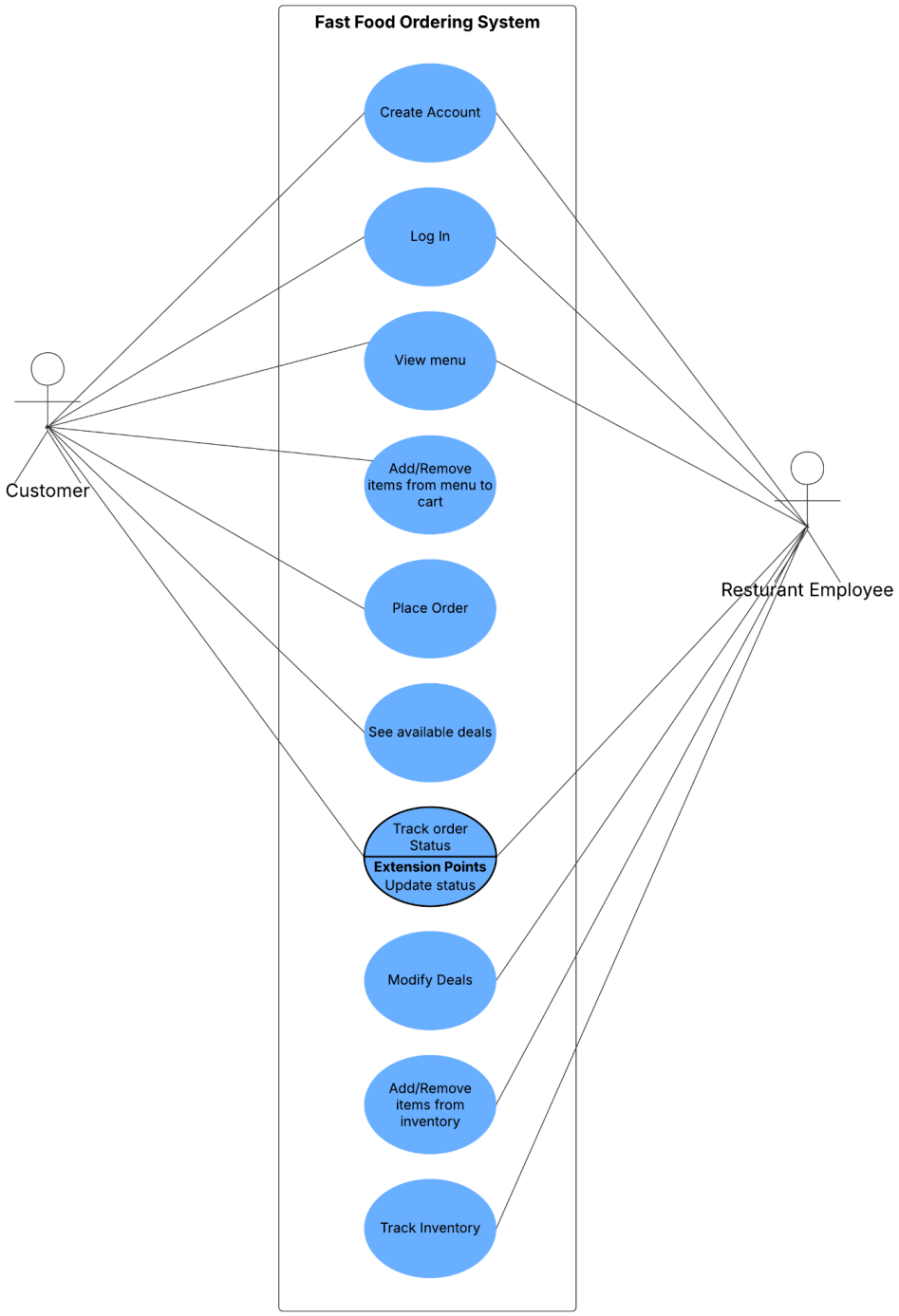
2.1.1 System Interfaces

The system interfaces are for Windows, Mac, and Linux, as well as Android or iPhone mobile devices.

2.1.2 User Interfaces

The user interface will be a web-based online ordering portal for efficient use by both customers and staff. The interface will be built using HTML and CSS, which is going to provide the structure and layout format, with JavaScript to allow for user interactivity.

2.1.2.1 Use Case Diagram



2.1.3 Hardware Interfaces

The user will be able to interface with the software through use of a mobile device or through a personal computer.

2.1.4 Software Interfaces

The user will be able to interface with the software through a web browser on their own personal device (Ex. Chrome, Firefox, Ect…)

2.1.5 Communications Interfaces

No specific communication interfaces are required. The client and server will communicate through standard network protocols.

2.1.6 Memory Constraints

This application will utilize garbage collecting languages, such as JS and Python, so explicit memory management most likely won't be needed.

2.1.7 Operations

The system will operate as a web app through modern web browsers such as Chrome and Edge. All operations are defined in section 3.

2.2 Product Functions

The system's primary functions will include:
Customer privileges:

- Browning Menu
- Editing their cart
- Placing orders
- Receiving alerts about inventory availability and order updates

Staff privileges:

- Staff login
- Accessing staff dashboard
- Accessing the order manager
- Accessing the inventory tracker

2.3 User Characteristics

There are two primary users for this app.
Customers:

- This is the target user for this application. This user's goal is to browse the menu and receive food.
- This user is typically someone who likes to order their food online without the hassle of going in person

Staff Member:

- This user's goal is to serve the customer
- This user works for the company that distributes the food
- This user will have access to the privileges granted after using the Staff Login and must use those privileges responsibly

2.4 Constraints

2.4.1 No Budget

There is no budget assigned to the project due to it being a class project without any intention of real world application.

2.5 Assumptions and Dependencies

This application will rely on a stable internet connection, access to a device that supports the formatting of the application, and a valid web browser. The application will only function properly if backend services and APIs are properly set up. The hosting environment must support all languages used to program the application.

2.6 Apportioning of Requirements

All requirements specified must be properly implemented by the planned delivery dates. Additional requirements must be well thought out in order to integrate them properly by the end of the timeline.
If something happens that will critically impact the predicted delivery time, the timeline and descriptions will be adjusted accordingly.

3 Specific Requirements

3.1 External Interface Requirements

- The system will use clear tabs and buttons so users can easily move between pages.
- Customers will be able to view the menu, add items to a cart, place orders, and track their order status.
- The Employee's interface will give staff access to the Order Manager and Inventory Tracker.
- The system shall make employee-only features visible to only logged-in employees.
- The system will use an interface that works with modern web browsers.
- The system will provide clear and visible feedback for possible invalid inputs such as login errors or empty cart messages.

3.2 Functional Requirements (by system feature)

Menu and Cart

- The system will allow customers to add items to a cart.
- Display all available menu items along with their prices so customers can easily make a selection.
- Allows customers to remove or update items in the cart before placing an order.
- Allow Customers to make specific customizations to items before placing an order.

- The system will display highlighted items and discounts from the menu on the landing page, labeled as “HOT” and “DEALS.”

Order placement and Tracking

- Allows customers to place an order after confirming their cart.
- It checks inventory before confirming the order.
- Customers can track their order status in real time.
- System shall generate a unique order ID for the order, if ordering is successful.
- If ordering is not successful, system should notify the customer and suggest other options.
- System should provide a receipt to the customer when a order is placed.

Order Manager

- Gives staff access to the Order Manager.
- Employees will be able to view all incoming and active customer orders in one place.
- Employees will be able to update statuses as orders move through the preparation process.
- System should allow employees to search for orders using their unique order ID.
- The system should remove completed orders from active orders when an order is finished.

Inventory Tracker

- The employee interface will give staff access to the inventory tracker.
- It will track item quantities in inventory.
- Inventory quantities will update automatically when an order is placed.
- The system should reject the ordering of items that are marked out of stock.
- The system should provide feedback to employees on rejected orders due to being out of stock.

3.3 Non-Functional Requirements

Performance

- Users should be able to travel across the system's pages in 3 seconds or less under normal circumstances.
- Users' orders should be processed and placed by the system within 10 seconds of pressing “Pay & Order."
- After ordering, the system should display a receipt for that order to the customer within 5 seconds.

Security

- Employee passwords should be at least 8 characters long.
- Employee passwords should contain at least 2 special characters, at least 1 uppercase letter, and at least 1 number.

Usability

- Users should be able to access the full menu within only 1 click.
- Users should be able to consistently place an average-sized order within 5 minutes.
- Employees should be able to search and find an order using order ID in the Order Manager in 10 seconds.
- Buttons and input fields should have descriptive labels.
- Readability: Page fonts should be a standard and easy-to-read font with a font size of 12px.

3.3.1 Space Requirement

- Stores order, inventory, and employee data in a database.
- Will use an appropriate amount of storage to maintain the data and application.

3.3.2 Time Requirements

- The system will update order statuses shortly after employees make changes.
- Pages such as the menu and cart should load without much delay.

3.4 Logical Database Requirements

- Stores customer orders and each order's status through an ID.
- Stores inventory items and quantities.
- Stores employees' login information securely and safely.

3.5 Design Constraints

- Constrained to the languages used, such as HTML, CSS, JavaScript, Python, and backend APIs.
- Limited to the scope and time provided for the class project.

3.6 Software System Attributes

- The system must be easy to maintain and update.
- Support multiple users at the same time.

4 Supporting Information

4.1 Appendices

4.1.1 Incremental Development Plan

Feature	Phase	Task	Description	Technologies
Accounts	1	Create Account	Allow users to create accounts, and specify whether they are a customer or employee, allowing different permissions	Javascript, SQL
Accounts	3	Update Account	Allow the user to update account information or delete the account	Javascript, SQL
Menu	1	Add items to the menu	Allow employees to add items to the menu	Javascript, SQL, HTML
Menu	1	Remove items from the menu	Allow employees to remove items from the menu	Javascript, SQL, HTML
Menu	3	Add deals to the menu	Allow for deals to be shown on specific menu items	Javascript, SQL, HTML
Cart	1	Add items to cart	Allow customers to add items to their individual cart	Javascript, HTML
Cart	1	Remove items from cart	Allow customers to remove items from their individual cart	Javascript, HTML
Orders	2	Place order	Allow customers to place an order with the items in their cart	Javascript, HTML
Orders	2	Track order	Allow customers to track the status of their order	Javascript
Orders	3	Search for Order	Allow employees to search orders by ID	SQL
Orders	2	Update Order	Allow employees to change the status of orders	SQL
Inventory	2	Track Inventory	Allow employees to view what is currently in the inventory of the restaurant	HTML

4.1.2 Team Contributions

Member Name	Contribution Description	Overall Contribution (%)	Notes

Kyle Burchett	UML Use Case Diagram / Development plan layout	16.67%	
Alena Emmanuelli	Overall Description	16.67%	
Cody Howell	Specific Requirements,	16.67%	(Some Functional Requirements, Non-Functional Requirements)
Jayden Patton	Overall description	16.67%	
Taying Yang (Tai Yang)	Introduction Descriptions	16.67%	
Kenny Lee	Specific Requirements	16.67%	